

Do Genome-Based Antibiotic Resistance Predictions Hold Up Over Time? A Date-Split Benchmark, Error Anatomy, and Structural Early-Warning Test on 12,841 Public Isolates

amr-genome-early-warning - research report

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- [1 Abstract](#)
- [2 Introduction](#)
 - [2.1 The problem](#)
 - [2.2 Why time matters](#)
 - [2.3 Research questions](#)
- [3 Background and prior art](#)
 - [3.1 Rule-based genotype-to-phenotype prediction](#)
 - [3.2 Machine-learning prediction](#)
 - [3.3 Novel-variant and surveillance approaches](#)
 - [3.4 Closest prior work: temporal validation](#)
 - [3.5 Gap addressed here](#)
- [4 Data](#)
 - [4.1 Sources](#)
 - [4.2 Inclusion](#)
 - [4.3 Test set sizes \(Gate G0\)](#)
- [5 Methods](#)
 - [5.1 Pre-registration](#)
 - [5.2 Features](#)
 - [5.3 Models](#)
 - [5.4 Splits and statistics](#)
 - [5.5 Error anatomy \(G2\)](#)
 - [5.6 Novelty score \(G3\)](#)
 - [5.7 Structural layer \(G4\)](#)
- [6 Results](#)
 - [6.1 RQ1: accuracy falls on newer isolates \(G1 PASS\)](#)
 - [6.1.1 Date-split detail for the random forest](#)
 - [6.1.2 Honest negative on the rule baseline](#)
 - [6.2 RQ2: new-era misses are mostly known genes \(G2\)](#)
 - [6.3 RQ3: a novelty score does not flag misses \(G3 FAIL, negative kept\)](#)
 - [6.4 RQ4: structural analysis of new beta-lactamase alleles \(G4, descriptive\)](#)
- [7 The tool: amr-ew](#)
- [8 Discussion](#)
 - [8.1 What this means](#)
 - [8.2 Comparison with prior art](#)
 - [8.3 Limitations](#)
 - [8.4 Future work](#)
- [9 Conclusion](#)
- [10 Reproducibility](#)
- [11 References](#)

- [12 Appendix A: all 60 model-split cells](#)
- [13 Appendix B: error anatomy by drug and species](#)
- [14 Appendix C: known determinants most often present in missed resistant isolates](#)
- [15 Appendix D: gate ledger](#)

1 Abstract

Problem. Tools that predict antibiotic resistance from a bacterial genome are usually tested on a random slice of the same collection they were built from. Hospitals, however, use them on isolates collected *after* the tool was built. If resistance biology, testing practice, or the mix of strains changes over time, random-split accuracy will overstate real-world accuracy.

Data. We assembled 12,841 *Klebsiella pneumoniae* and *Escherichia coli* isolates from NCBI Pathogen Detection that have (i) an AMRFinderPlus genotype, (ii) a laboratory antimicrobial susceptibility test (AST) result for at least one of five drugs, and (iii) a collection year. We added the AMRFinderPlus reference catalog (database 2026-08-07.1), allele protein sequences, and four beta-lactamase crystal structures from RCSB PDB.

Methods. Gates were locked in a standalone commit before any model was run. We compared three predictors (a class-level rule, logistic regression, random forest) under a random split and a date split (train on isolates collected up to 2019, test on 2020 onward), with bootstrap confidence intervals. We then classified every date-split miss by cause, tested whether a “novelty” score flags misses, and mapped new beta-lactamase alleles onto protein structures.

Results. (1) Accuracy drops on newer isolates in all 20 machine-learning cells: mean balanced accuracy (BA) falls by 0.134 (95% CI 0.109-0.158). The largest drop is ceftazidime in *K. pneumoniae* (random forest, 0.900 to 0.666). (2) Of 1,281 resistant isolates the random forest missed on the date split, only 41 (3.2%) carried a resistance gene the model had never seen; 869 (67.8%) carried a *known* determinant of that drug class, and 371 (29.0%) had no determinant at all. (3) **Negative result:** a pre-registered novelty score did not flag misses (AUROC 0.52, CI 0.51-0.54; gate needed 0.65). (4) **Negative result:** newer beta-lactamase alleles were not enriched for substitutions near the catalytic site (odds ratio 0.80, $p = 0.54$, exploratory).

Tool. From these results we built `amr-ew`, a command-line predictor that reports its *date-split* accuracy instead of the optimistic random-split number, adds per-determinant early-warning flags based on which known genes are missed most often in new isolates, and annotates beta-lactamase alleles with structural distance to the active site.

Contribution. A cross-species, date-split benchmark showing that the main reason predictions fail on new isolates is a changed genotype-to-phenotype link for *known* genes, not new genes. This moves the early-warning target away from “watch for new genes” toward “watch for known genes that stop behaving as expected”.

2 Introduction

2.1 The problem

Antimicrobial resistance (AMR) is one of the most serious global health threats. The AMRscope authors summarise estimates of about 700,000 deaths per year, projected to rise sharply by 2050 (AMRscope, bioRxiv 2025 [7]). Doctors often must start antibiotics before laboratory susceptibility results are ready, so fast genome-based prediction could guide the first choice of drug.

Two families of methods exist. **Rule-based** tools such as ResFinder 4.0 [1] and NCBI AMRFinderPlus [2] detect known resistance genes and point mutations and translate them into predicted phenotypes. **Machine-learning (ML)** methods learn the link between genome features and laboratory phenotype from large collections [3, 4, 5].

2.2 Why time matters

Most published evaluations split one collection randomly into training and test sets. Because isolates from the same outbreak, lab and year land on both sides, the test set looks very much like the training set. A deployed tool meets isolates from later years, possibly with new genes, new strain backgrounds, and changed lab breakpoints. The question of this project is simple: **how much accuracy is lost when a genome-based AMR predictor is tested on isolates collected after its training data, and why?**

2.3 Research questions

1. **RQ1 (G1).** Is date-split accuracy lower than random-split accuracy, across species, drugs and models?
2. **RQ2 (G2).** When the model misses a resistant isolate from the new era, is it because of an unseen gene, a known gene, or no detectable gene?
3. **RQ3 (G3).** Can a simple novelty score flag those misses early?
4. **RQ4 (G4).** Do newly appearing beta-lactamase alleles carry substitutions close to the enzyme's active site?
5. **RQ5 (G5).** Can these findings be turned into a practical tool?

3 Background and prior art

3.1 Rule-based genotype-to-phenotype prediction

ResFinder 4.0 added phenotype prediction from detected genes and point mutations for several species [1]. The AMRFinder validation study compared tool calls with measured phenotypes on a fixed collection of isolates [2]. A systematic review of AMR marker databases found performance varies widely by drug and species [3]. All of these evaluate on fixed collections, not on isolates collected after the database was built.

3.2 Machine-learning prediction

ML methods predict phenotype from genes, SNPs or k-mers. Recent examples include *E. coli* WGS classifiers trained on 1,952 isolates [4], work on *Acinetobacter baumannii* feature limits [5], and interpretable deep learning for *K. pneumoniae* [6]. VAMPr maps variants to resistance with explainable features [8]. A large community benchmark assessed computational AMR phenotype predictions from genomes [9].

3.3 Novel-variant and surveillance approaches

AMRscope uses protein language model embeddings to score single amino-acid substitutions for resistance risk and tests generalisation with gene, organism and homology hold-outs, but not date hold-outs [7]. DRAMMA detects novel AMR genes in metagenomes [10]. amr.watch monitors AMR trends from global genomic data but does not benchmark phenotype prediction [11]. A PLOS Biology study argued that genotype-based diagnostics need ongoing surveillance to stay sensitive [12].

3.4 Closest prior work: temporal validation

The open-source project kleb-amr-project trains *K. pneumoniae* models on pre-2023 isolates and tests on 2023-2024 [13]. That is a single-species temporal test. It does not break down *why* new-era isolates are missed, and it does not test an early-warning signal.

3.5 Gap addressed here

Aspect	Rule tools [1,2]	AMRscope [7]	amr.watch [11]	kleb-amr-project [13]	This work
Phenotype prediction	yes	variant risk	no	yes	yes
Date-split test	no	no	n/a	yes (1 species)	yes (2 species, 5 drugs, 3 models)
Cause of new-era misses	no	no	no	no	yes
Early-warning score tested	no	partly	trends	no	yes (negative)
Structural mapping of new alleles	no	AlphaFold heatmaps	no	no	yes

4 Data

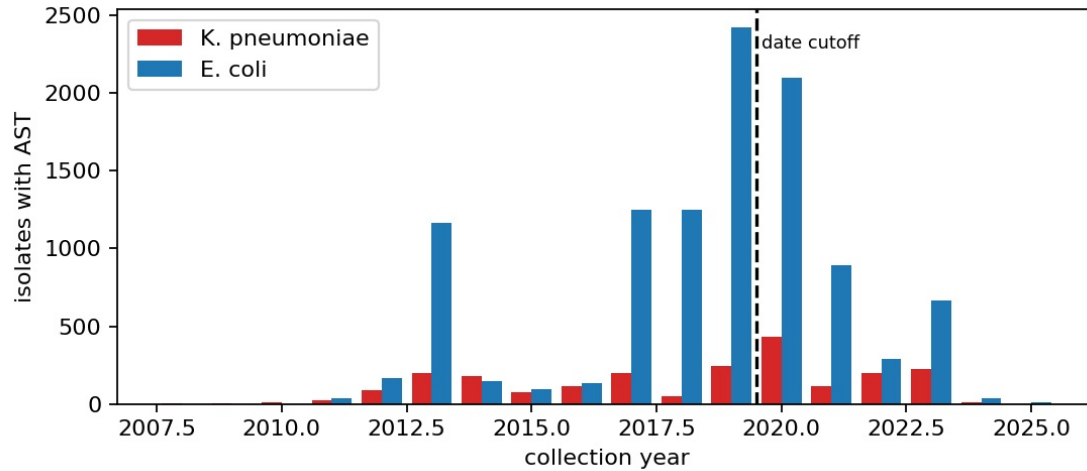
4.1 Sources

Dataset	Content	Source
NCBI Pathogen Detection PDG000000012.2532	<i>K. pneumoniae</i> isolates, genotypes, AST	ftp.ncbi.nlm.nih.gov/pathogen/Results/Klebsiella
NCBI Pathogen Detection PDG000000004.6316	<i>E. coli/Shigella</i> isolates	ftp.ncbi.nlm.nih.gov/pathogen/Results/Escherichia_coli_Shigella
AMRFinderPlus DB 2026-08-07.1	ReferenceGeneCatalog, AMRProt.fa, mutation tables	ftp.ncbi.nlm.nih.gov/pathogen/Antimicrobial_resistance/AMRFinderPlus
RCSB PDB	KPC-2 (2OV5), SHV-1 (1SHV), NDM-1 (4EYL), IMP-1 (1DD6)	files.rcsb.org

Every file’s URL and SHA-256 checksum is listed in data/SOURCES.md. The NCBI AST data are described in the NCBI Insights post on the AST browser [14].

4.2 Inclusion

An isolate was included if it had an AST result (R/I/S) for at least one of ceftazidime, meropenem, ciprofloxacin, gentamicin, or trimethoprim-sulfamethoxazole, and a four-digit collection year. This gave **12,841 isolates**. Intermediate (I) results were excluded from the primary analysis.



Isolates with AST results by collection year. The dashed line marks the date-split cutoff (train up to 2019, test 2020 onward).

4.3 Test set sizes (Gate G0)

Drug	Species	Train n (R)	Test n (R)
ceftazidime	K. pneumoniae	984 (537)	733 (224)
ceftazidime	E. coli	1114 (348)	2487 (593)
meropenem	K. pneumoniae	605 (314)	486 (143)
meropenem	E. coli	3921 (78)	2602 (108)
ciprofloxacin	K. pneumoniae	576 (342)	518 (189)
ciprofloxacin	E. coli	5152 (515)	2854 (946)
gentamicin	K. pneumoniae	576 (283)	462 (194)
gentamicin	E. coli	5077 (892)	2602 (380)
trimethoprim-sulfamethoxazole	K. pneumoniae	558 (333)	521 (197)
trimethoprim-sulfamethoxazole	E. coli	5037 (716)	2385 (695)

All ten drug-species cells passed G0 (at least 100 test isolates and 20 resistant).

5 Methods

5.1 Pre-registration

The analysis plan and pass/fail thresholds were committed as GATES_LOCKED.md (commit 9f770aa) before any model was trained. Before that commit, raw metadata were downloaded and only counted (isolates per year, tests per drug) to check feasibility; this is disclosed in DEVIATIONS.md. No gate threshold was changed after results were seen.

5.2 Features

Each isolate is a presence/absence vector of AMRFinderPlus determinants (genes and point mutations) listed in the AMR_genotypes field. The vocabulary is built from the training set only, so a determinant first seen after the cutoff is invisible to the model.

5.3 Models

- **M0, class-level rule:** predict R if the isolate carries any determinant whose AMRFinderPlus subclass matches the drug's class (both a trimethoprim and a sulfonamide determinant for trimethoprim-sulfamethoxazole).
- **M1:** L2-regularised logistic regression.
- **M2:** random forest, 500 trees, seed 42.

5.4 Splits and statistics

- **Random split:** stratified 80/20.
- **Date split:** train on collection year up to 2019, test on 2020 onward.
- Primary metric: balanced accuracy (BA), the mean of sensitivity and specificity. Secondary: Matthews correlation coefficient (MCC) and sensitivity for resistance.
- 95% confidence intervals from 1,000 bootstrap resamples.
- G1 passes if the mean BA drop across ML cells is at least 0.02 and its CI excludes zero.

5.5 Error anatomy (G2)

Each date-split false negative (resistant isolate called susceptible by M2) was assigned to one category: (a) carries a determinant of the drug's class that was absent from the training vocabulary; (b) carries only known determinants of the class; (c) carries no determinant of the class.

5.6 Novelty score (G3)

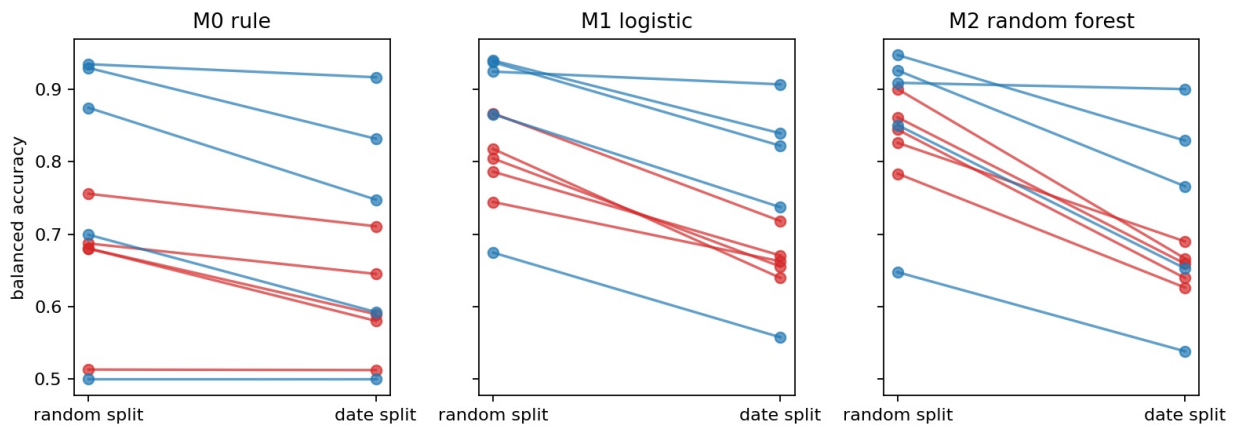
Score = number of determinants unseen before the cutoff + 2 x number of unseen determinants of the drug's class. Gate: AUROC for separating missed from caught resistant isolates of at least 0.65 with CI lower bound above 0.5.

5.7 Structural layer (G4)

For four beta-lactamase families, each allele's protein sequence (AMRProt.fa) was aligned (Biopython, BLOSUM62) to its family reference (KPC-2, SHV-1, NDM-1, IMP-1), and the reference to the PDB chain. For every substitution we computed the minimum heavy-atom distance from the substituted residue to the catalytic site: the Ser70 side-chain oxygen for class A enzymes (verified as residue SER 70 in both structures) or any zinc ion for metallo-beta-lactamases. "Proximal" means 8 Angstrom or closer. An allele's first year is the earliest collection year it appears in the two PDG snapshots.

6 Results

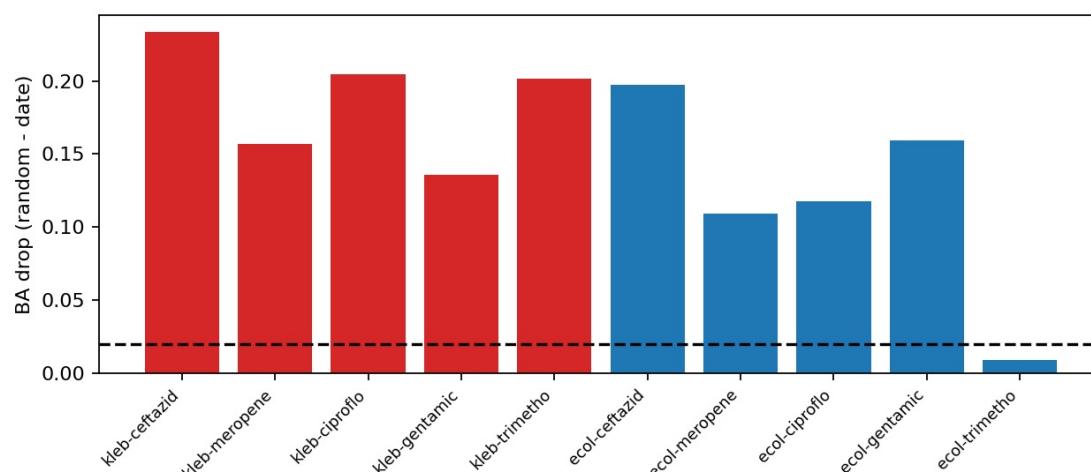
6.1 RQ1: accuracy falls on newer isolates (G1 PASS)



Balanced accuracy of each drug-species cell under random and date splits. Red = *K. pneumoniae*, blue = *E. coli*.

Across the 20 ML cells the mean BA drop was **0.134 (95% CI 0.109-0.158)**. Every cell dropped. The near-null exception was trimethoprim-sulfamethoxazole in *E. coli* (random forest drop 0.009), where accuracy stayed near 0.90.

Species	Drug	Model	BA random	BA date	Drop
<i>K. pneumoniae</i>	ceftazidime	M1	0.867	0.718	+0.148
<i>K. pneumoniae</i>	ceftazidime	M2	0.900	0.666	+0.233
<i>K. pneumoniae</i>	meropenem	M1	0.744	0.663	+0.082
<i>K. pneumoniae</i>	meropenem	M2	0.783	0.626	+0.157
<i>K. pneumoniae</i>	ciprofloxacin	M1	0.817	0.640	+0.178
<i>K. pneumoniae</i>	ciprofloxacin	M2	0.844	0.640	+0.204
<i>K. pneumoniae</i>	gentamicin	M1	0.804	0.655	+0.149
<i>K. pneumoniae</i>	gentamicin	M2	0.826	0.690	+0.136
<i>K. pneumoniae</i>	TMP-SMX	M1	0.786	0.671	+0.116
<i>K. pneumoniae</i>	TMP-SMX	M2	0.861	0.659	+0.202
<i>E. coli</i>	ceftazidime	M1	0.865	0.737	+0.128
<i>E. coli</i>	ceftazidime	M2	0.850	0.653	+0.197
<i>E. coli</i>	meropenem	M1	0.674	0.558	+0.117
<i>E. coli</i>	meropenem	M2	0.647	0.538	+0.109
<i>E. coli</i>	ciprofloxacin	M1	0.939	0.839	+0.100
<i>E. coli</i>	ciprofloxacin	M2	0.947	0.829	+0.118
<i>E. coli</i>	gentamicin	M1	0.937	0.822	+0.115
<i>E. coli</i>	gentamicin	M2	0.926	0.766	+0.160
<i>E. coli</i>	TMP-SMX	M1	0.924	0.907	+0.018
<i>E. coli</i>	TMP-SMX	M2	0.909	0.900	+0.009

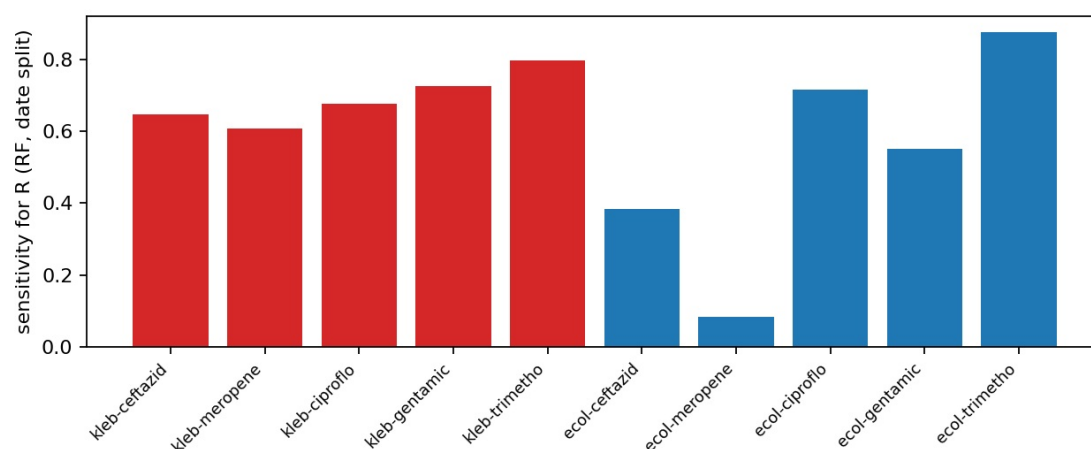


Random forest BA drop per cell. Dashed line = the 0.02 gate threshold.

A notable pattern: the random forest, which looks best on random splits, often loses the most on the date split. In *K. pneumoniae* ceftazidime, logistic regression beats the forest on new isolates (0.718 vs 0.666). More flexible models appear to fit period-specific patterns.

6.1.1 Date-split detail for the random forest

Species	Drug	BA (95% CI)	MCC	Sensitivity for R
K. pneumoniae	ceftazidime	0.666 (0.629-0.701)	0.311	0.647
K. pneumoniae	meropenem	0.626 (0.578-0.672)	0.233	0.608
K. pneumoniae	ciprofloxacin	0.640 (0.595-0.680)	0.269	0.677
K. pneumoniae	gentamicin	0.690 (0.649-0.732)	0.375	0.727
K. pneumoniae	TMP-SMX	0.659 (0.618-0.697)	0.315	0.797
E. coli	ceftazidime	0.653 (0.634-0.672)	0.365	0.383
E. coli	meropenem	0.538 (0.514-0.568)	0.157	0.083
E. coli	ciprofloxacin	0.829 (0.813-0.845)	0.694	0.715
E. coli	gentamicin	0.766 (0.742-0.793)	0.639	0.550
E. coli	TMP-SMX	0.900 (0.885-0.913)	0.786	0.876



Sensitivity for resistance on 2020+ isolates (random forest). *E. coli* meropenem resistance is almost never detected (8%).

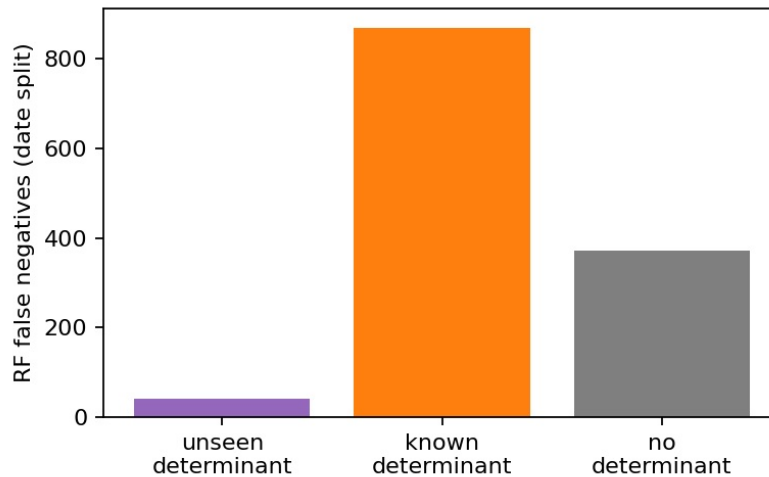
The weakest cell is *E. coli* meropenem: only 78 resistant isolates in training, and sensitivity of 8% on new isolates. A carbapenem-resistant *E. coli* would usually be called susceptible, the most dangerous

error.

6.1.2 Honest negative on the rule baseline

The locked class-level rule (M0) collapsed to BA near 0.50 for *E. coli* ceftazidime and *K. pneumoniae* ciprofloxacin. The reason is that intrinsic chromosomal genes (such as *blaEC* in *E. coli* and *oqxAB* in *K. pneumoniae*) carry the drug-class label in the catalog, so the rule calls nearly everything resistant. We kept the rule as locked and report this: class-level rules that ignore intrinsic genes are not a fair baseline for these cells, and practical rule tools [1, 2] use curated drug-specific logic.

6.2 RQ2: new-era misses are mostly known genes (G2)



Cause of each random-forest false negative on the date split ($n = 1,281$).

Category	Count	Share
(a) unseen determinant of the drug class	41	3.2%
(b) known determinant of the drug class	869	67.8%
(c) no determinant of the drug class	371	29.0%

A concern is that intrinsic chromosomal genes (for example *blaEC* in *E. coli*) carry drug-class labels and could inflate category (b). In a **post-hoc sensitivity analysis** (not pre-registered) we removed determinants present in 80% or more of a species' isolates (*emrD*, *fosA*, *oqxA* in *K. pneumoniae*; *acrF*, *blaEC* in *E. coli*). Only 25 isolates moved from (b) to (c): 41 / 844 / 396. Category (b) remains 65.9% of misses.

This is the main finding. **New genes are not the main problem.** Two thirds of the missed resistant isolates carry genes the model had already seen. The model had learned from pre-2020 data that those genes were not enough to predict resistance, but in newer isolates they were. Possible reasons include changes in the genetic background, gene expression or copy number, lab methods, or the breakpoints used to call R and S. Our data cannot separate these; the per-isolate testing standard was not in the metadata we used. The 29% with no determinant point to mechanisms outside the AMRFinderPlus catalog (for example porin loss or efflux upregulation without a catalogued mutation).

6.3 RQ3: a novelty score does not flag misses (G3 FAIL, negative kept)

The novelty score gave **AUROC 0.522 (95% CI 0.507-0.536)** across 3,669 resistant test isolates (1,281 missed). The gate needed 0.65. This follows from RQ2: if only 3% of misses involve unseen

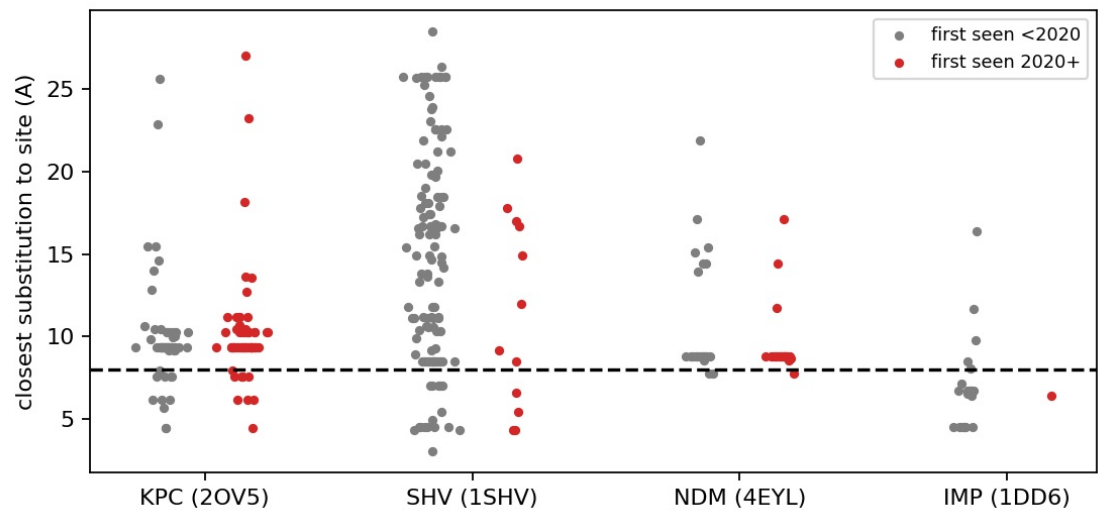
genes, counting unseen genes cannot find the other 97%. We did not re-tune the score.

6.4 RQ4: structural analysis of new beta-lactamase alleles (G4, descriptive)

Only five alleles first seen in 2020 or later appear in date-test isolates, below the n of 10 needed for the locked Fisher test, so they are described:

Allele	First year in PDG	Substitutions vs reference (distance to site, A)	Closest (A)	In a missed isolate
blaKPC-45	2020	T92K (23.2)	23.2	no
blaSHV-57	2024	L165R (6.6)	6.6	no
blaSHV-250	2022	Y3F (signal peptide), G232E (5.4)	5.4	no
blaNDM-50	2022	M22I (signal region), D130N (17.1), M154L (8.8)	8.8	yes
blaIMP-18	2020	48 substitutions (distant IMP homolog)	6.4	no

“First year” means first appearance in this surveillance stream, not the year of first publication; some alleles were described earlier (see DEVIATIONS.md).



Closest substitution to the catalytic site for every allele with a substitution, by family and first-seen period. Dashed line = 8 A.

Exploratory (not pre-registered): across all 338 alleles with substitutions in the four families, 16 of 101 post-2019 alleles and 45 of 237 earlier alleles had a substitution within 8 A of the site (OR 0.80, $p = 0.54$). **Negative:** newer alleles are not more active-site-focused. KPC is the fastest-diversifying family in this stream: 64 of 125 KPC alleles were first seen in 2020 or later.

7 The tool: amr-ew

Each design choice in the tool comes from a result:

Result	Design choice in amr-ew
G1: random-split accuracy is optimistic by 0.13 BA	Each call reports the model's <i>date-split</i> BA for that drug and species
G2: 68% of misses carry known genes	Early-warning flag when a carried determinant was missed in 30% or more of resistant 2020+ isolates
G3: novelty does not predict misses	Unseen determinants are listed as information only, never as a confidence score
G4: structural distance is informative per allele but not a group signal	Per-allele distance to active site reported for KPC, SHV, NDM, IMP

Example:

```
python3 tool/amr_ew.py --species klebsiella --year 2023 \
  --genotypes "blaKPC-2,blaSHV-11,gyrA_S83I,parC_S80I"
```

Output (JSON) gives, per drug: R/S call, probability, honest date-split BA, and early-warning messages.

Validation (G5). On 30 random 2020+ isolates (seed 42) the tool made 77 drug calls with 0.870 accuracy. All 7 missed resistant calls carried an early-warning flag, but so did 6 of 9 correct resistant calls. The flags were derived on the same test period, so this is in-sample and optimistic. The flags are sensitive but not specific. We report this as a limitation, not a validated early-warning claim.

8 Discussion

8.1 What this means

A genome-based AMR model that reports 0.90 balanced accuracy on a random split may deliver about 0.67 on isolates from the following years. Anyone deploying such a model should ask for date-split numbers. Our benchmark gives a reusable way to produce them from public NCBI data.

The error anatomy changes where early warning should look. Surveillance systems often focus on new genes and new alleles [10, 11]. In our data those explain only 3% of misses. The larger signal is a known gene whose link to phenotype changes. A practical early-warning system should track, for each known determinant, the rate at which carriers test resistant, and alarm when that rate shifts. This matches the argument that genotype-based diagnostics need continuous surveillance to stay sensitive [12].

8.2 Comparison with prior art

- Versus **kleb-amr-project** [13]: we extend temporal validation to two species and three model types, and add the cause-of-miss breakdown.
- Versus **AMRscope** [7]: AMRscope's hold-outs are by gene, organism and homology; ours is by date, the axis a deployed tool actually faces. Our structural negative suggests that position near the active site alone does not mark newer alleles.
- Versus **rule tools** [1, 2]: our naive class rule failed where intrinsic genes exist, a reminder that rule tools depend on careful drug-specific curation.

Quantified methodological contribution. On identical isolates and features, a random-split benchmark overstates new-isolate balanced accuracy by 0.134 on average (up to 0.233). The error decomposition shows that 96.8% of new-era misses are not addressable by novelty detection, the approach behind novel-gene early-warning tools.

8.3 Limitations

1. Isolates in NCBI Pathogen Detection with AST are not a random sample of infections; many come from US surveillance and a few large projects.
2. We used AMRFinderPlus calls as features and did not re-assemble genomes; SNP and k-mer features might behave differently.
3. The cause of the known-gene shift (breakpoints, labs, strain background) could not be separated with the metadata used.
4. Only one cutoff (2019/2020) was tested, as locked.
5. The tool's early-warning flags are validated in-sample only.
6. Structural analysis used crystal structures of reference enzymes, not models of each allele.

8.4 Future work

Add the per-isolate testing standard to test the breakpoint explanation; add rolling cutoffs; include AlphaFold models per allele; and test determinant-level drift alarms prospectively on isolates released after this snapshot.

9 Conclusion

Genome-based AMR prediction loses substantial accuracy on newer isolates in every drug and species tested. The loss comes mainly from known resistance genes whose relation to the measured phenotype shifts, not from new genes. Two pre-registered ideas failed and are reported: a novelty score does not flag misses, and newer beta-lactamase alleles are not enriched near the active site. The `amr-ew` tool turns these results into honest per-drug accuracy reporting and determinant-level warnings.

10 Reproducibility

- Code: `code/build_dataset.py`, `code/analysis.py`, `code/structural_layer.py`, `code/figures.py`, `tool/amr_ew.py`, `tool/validate_tool.py`.
- Results: `results/` (JSON, TSV, markdown gate reports).
- Gates: `GATES_LOCKED.md` (commit 9f770aa); deviations: `DEVIATIONS.md`.
- Every committed file is listed with its SHA-256 in `MANIFEST.sha256`.
- Software: Python 3, scikit-learn 1.7.2, Biopython 1.88, SciPy, matplotlib. Random seed 42 throughout.

11 References

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12 Appendix A: all 60 model-split cells

Species	Drug	Split	Model	n test	n R	BA (95% CI)	MCC	Sens.
klebsiella	ceftazidime	random	M0	344	152	0.756 (0.716-0.794)	0.539	0.954
klebsiella	ceftazidime	random	M1	344	152	0.867 (0.828-0.903)	0.730	0.868
klebsiella	ceftazidime	random	M2	344	152	0.900 (0.864-0.928)	0.796	0.914
klebsiella	ceftazidime	date	M0	733	224	0.711 (0.683-0.741)	0.394	0.879
klebsiella	ceftazidime	date	M1	733	224	0.718 (0.683-0.751)	0.410	0.705
klebsiella	ceftazidime	date	M2	733	224	0.666 (0.629-0.701)	0.311	0.647
klebsiella	meropenem	random	M0	219	92	0.680 (0.621-0.738)	0.361	0.793
klebsiella	meropenem	random	M1	219	92	0.744 (0.686-0.800)	0.507	0.630
klebsiella	meropenem	random	M2	219	92	0.783 (0.728-0.835)	0.584	0.685
klebsiella	meropenem	date	M0	486	143	0.590 (0.538-0.638)	0.163	0.622
klebsiella	meropenem	date	M1	486	143	0.663 (0.618-0.709)	0.317	0.552
klebsiella	meropenem	date	M2	486	143	0.626 (0.578-0.672)	0.233	0.608
klebsiella	ciprofloxacin	random	M0	219	106	0.513 (0.500-0.529)	0.114	1.000
klebsiella	ciprofloxacin	random	M1	219	106	0.817 (0.767-0.868)	0.635	0.821
klebsiella	ciprofloxacin	random	M2	219	106	0.844 (0.794-0.891)	0.689	0.821
klebsiella	ciprofloxacin	date	M0	518	189	0.513 (0.502-	0.084	0.995

						0.523)		
klebsiella	ciprofloxacin	date	M1	518	189	0.640 (0.597- 0.684)	0.269	0.656
klebsiella	ciprofloxacin	date	M2	518	189	0.640 (0.595- 0.680)	0.269	0.677
klebsiella	gentamicin	random	M0	208	96	0.680 (0.616- 0.743)	0.359	0.708
klebsiella	gentamicin	random	M1	208	96	0.804 (0.749- 0.855)	0.612	0.760
klebsiella	gentamicin	random	M2	208	96	0.826 (0.769- 0.878)	0.664	0.750
klebsiella	gentamicin	date	M0	462	194	0.580 (0.534- 0.627)	0.159	0.582
klebsiella	gentamicin	date	M1	462	194	0.655 (0.613- 0.702)	0.307	0.706
klebsiella	gentamicin	date	M2	462	194	0.690 (0.649- 0.732)	0.375	0.727
klebsiella	trimethoprim- sulfamethoxazole	random	M0	216	106	0.687 (0.627- 0.746)	0.382	0.792
klebsiella	trimethoprim- sulfamethoxazole	random	M1	216	106	0.786 (0.731- 0.842)	0.576	0.736
klebsiella	trimethoprim- sulfamethoxazole	random	M2	216	106	0.861 (0.814- 0.903)	0.722	0.849
klebsiella	trimethoprim- sulfamethoxazole	date	M0	521	197	0.645 (0.604- 0.684)	0.285	0.756
klebsiella	trimethoprim- sulfamethoxazole	date	M1	521	197	0.671 (0.631- 0.710)	0.337	0.807
klebsiella	trimethoprim- sulfamethoxazole	date	M2	521	197	0.659 (0.618- 0.697)	0.315	0.797
ecoli	ceftazidime	random	M0	721	188	0.500 (0.500- 0.500)	0.000	1.000
ecoli	ceftazidime	random	M1	721	188	0.865 (0.835- 0.894)	0.738	0.793
ecoli	ceftazidime	random	M2	721	188	0.850 (0.819- 0.881)	0.717	0.761

ecoli	gentamicin	random	M2	1536	254	(0.903-0.946)	0.882	0.862
ecoli	gentamicin	date	M0	2602	380	0.832 (0.809-0.856)	0.726	0.684
ecoli	gentamicin	date	M1	2602	380	0.822 (0.800-0.845)	0.710	0.666
ecoli	gentamicin	date	M2	2602	380	0.766 (0.742-0.793)	0.639	0.550
ecoli	trimethoprim-sulfamethoxazole	random	M0	1485	282	0.934 (0.916-0.953)	0.869	0.894
ecoli	trimethoprim-sulfamethoxazole	random	M1	1485	282	0.924 (0.904-0.943)	0.856	0.872
ecoli	trimethoprim-sulfamethoxazole	random	M2	1485	282	0.909 (0.887-0.930)	0.831	0.844
ecoli	trimethoprim-sulfamethoxazole	date	M0	2385	695	0.916 (0.903-0.928)	0.823	0.894
ecoli	trimethoprim-sulfamethoxazole	date	M1	2385	695	0.907 (0.893-0.920)	0.801	0.883
ecoli	trimethoprim-sulfamethoxazole	date	M2	2385	695	0.900 (0.885-0.913)	0.786	0.876

13 Appendix B: error anatomy by drug and species

Species	Drug	(a) unseen	(b) known	(c) none	total
ecoli	ceftazidime	22	344	0	366
ecoli	ciprofloxacin	6	249	15	270
ecoli	gentamicin	1	66	104	171
ecoli	meropenem	2	10	87	99
ecoli	trimethoprim-sulfamethoxazole	1	59	26	86
klebsiella	ceftazidime	4	49	26	79
klebsiella	ciprofloxacin	3	57	1	61
klebsiella	gentamicin	0	2	51	53
klebsiella	meropenem	2	14	40	56
klebsiella	trimethoprim-sulfamethoxazole	0	19	21	40

14 Appendix C: known determinants most often present in missed resistant isolates

Share = missed resistant 2020+ carriers / all resistant 2020+ carriers (random forest, date split).
Determinants with at least 20 resistant carriers; top 30 by count of misses.

Species	Drug	Determinant	Missed	Resistant carriers	Share missed
ecoli	ceftazidime	acrF	355	579	0.61
ecoli	ceftazidime	blaEC	351	576	0.61
ecoli	ceftazidime	mdtM	324	531	0.61
ecoli	ciprofloxacin	acrF	267	929	0.29
ecoli	ciprofloxacin	blaEC	264	798	0.33
ecoli	ciprofloxacin	mdtM	244	720	0.34
ecoli	ceftazidime	sul2	189	284	0.67
ecoli	ceftazidime	tet(A)	176	281	0.63
ecoli	gentamicin	blaEC	165	362	0.46
ecoli	gentamicin	acrF	163	370	0.44
ecoli	gentamicin	mdtM	151	318	0.47
ecoli	ciprofloxacin	tet(A)	151	462	0.33
ecoli	ceftazidime	aph(6)-Id	136	208	0.65
ecoli	ceftazidime	aph(3'')-Ib	136	208	0.65
ecoli	ciprofloxacin	sul2	129	432	0.30
ecoli	ciprofloxacin	qnrS1	122	153	0.80
ecoli	ciprofloxacin	blaTEM-1	122	408	0.30
ecoli	ceftazidime	gyrA_S83L	122	263	0.46
ecoli	gentamicin	sul2	103	194	0.53
ecoli	ciprofloxacin	aph(6)-Id	103	392	0.26
ecoli	ceftazidime	blaTEM-1	103	171	0.60
ecoli	ceftazidime	aadA1	103	128	0.80
ecoli	ciprofloxacin	aph(3'')-Ib	102	391	0.26
ecoli	ciprofloxacin	floR	99	169	0.59
ecoli	meropenem	blaEC	98	107	0.92
ecoli	gentamicin	tet(A)	96	231	0.42
ecoli	meropenem	acrF	95	104	0.91
ecoli	gentamicin	gyrA_S83L	91	234	0.39
ecoli	ceftazidime	emrD	87	133	0.65
ecoli	meropenem	mdtM	85	93	0.91

Many top entries are intrinsic or near-universal genes (e.g. fosA, oqxA/B, blaSHV variants in *K. pneumoniae*); their presence in missed isolates is expected and not itself causal. Acquired determinants in this table are the candidates for determinant-level drift monitoring.

15 Appendix D: gate ledger

Gate	Criterion (locked)	Outcome
G0	$\geq 3,000$ isolates; each test cell ≥ 100 with ≥ 20 R	PASS (12,841; all 10 cells)
G1	mean ML BA drop ≥ 0.02 , CI excludes 0	PASS (0.134, CI 0.109-0.158)
G2	decompose all date-split FNs	DONE (41 / 869 / 371)
G3	novelty AUROC ≥ 0.65 , CI lower > 0.5	FAIL (0.522) - kept as negative
G4	Fisher test if $n \geq 10$, else descriptive	DESCRIPTIVE (n=5); exploratory null
G5	CLI runs on held-out date-split isolates	PASS (77 calls, accuracy 0.870)